

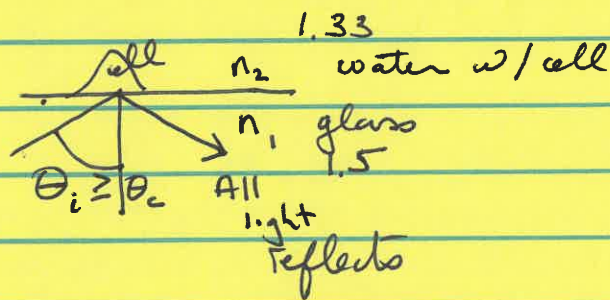
Lecture 21

3/9/26
JRA

HW is due Sunday at midnight
I may need to delete the last few questions
A.S.1 ; A.S.2 sign up will be available soon
PC, DF, & pol hands on

TIR Refresher

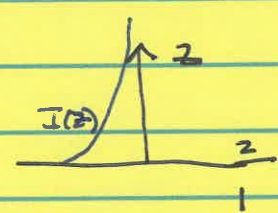
Light incident from region of higher " n "
(e.g. glass) onto region of lower " n " (e.g.,
watery cell) can undergo if incidence angle
 $\theta_i \geq \theta_c$ when $\sin \theta_c = n_2/n_1$.



This is geometrical optics view $\Rightarrow$ nothing gets into med 2
Wave optics predicts a (b.t of light) penetrates into medium 2 (evanescent)

Evanescent electric field penetrates into med 2

$$I = I_0 e^{-z/d}$$
 ↑ intensity in medium 2
 ↖ intensity at surface (right at interface)



$d \approx 100 \text{ nm}$	-1	0.3
	-2	0.135

Intensity basically gone w/in about about 200 nm

Ex
Light only excites Fluorescence right at interface

Key Ideas: (1) excitation light is essentially gone beyond $z \approx 200 \text{ nm}$

$\Rightarrow$ No Fluorescence collected beyond about 200 nm

(2) Blur free images because not exciting the out-of-focus fluorophores (fluorophores only exciting a thin layer)

Experimentally tell you have TIR because image changes from blurry to crisp

3) Reduced Photo B / photo T because not exciting most fluorophores

4) You know you're looking at the cell surface (allows observations of surface events)

d depends on $n, \lambda, \& \theta$

See formula in book

Application

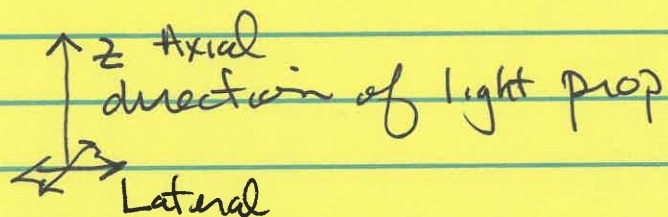
This mechanism is used in fiber optic cables



Lossless transmission along cable

Limitation: Restricted to cell surface

TIRFM is superresolution imaging in the z or axial direction



Diffraction-limited resolution in axial direction is $\sim 500-600$ nm

Lateral ~ 200 nm

TIR gives resolution of 100-200 nm

Significantly better in axial than LSCM

SUPER

There are 3 techniques that break resolution barrier ^{diff limit} in all 3 directions

Saw these in previous lecture

Discuss only superes SIM

Structured Illumination

100 nm lat / 300 nm axial

(1) 2X resolution improvement in all 3 directions

(2) Uses structured illumination (NOT ^{uniform} Kohler)
Key trick

Uses grid-like illumination



(3) Works w/ std fluorescent samples

Once you have fluorescent sample you're good
Other two techniques require special sample prep

(4) Well suited for multi color imaging
Can do w/ other techniques but it's more work

Looked at comparison of nuclear pore complex - much clearer than LSCM

Two techniques < Need special Fluorophores
than can be switched
between states

Here, all std Fluorophores are good

Theory complicated
Focus here on analogies

Std analogy for how SM works

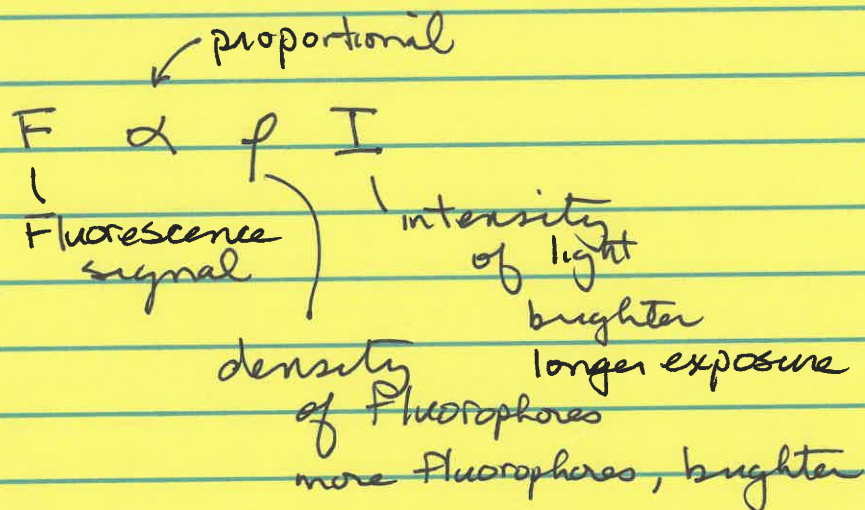
What does experimentalist want to know vs what gets measured?

Exp wants to know distribution of stuff in cells

Distribution or density of Fluorophores in the cell

What do you measure? Fluorescence signal

These two things are subtly different



IF I is uniform everywhere the same
 F and ρ just differ by a constant

$$F \propto \rho$$

This is why Kohler illum is good $\rightarrow$ reliably
 regard Fluor as indicative of ρ (i.e. NOT unwanted
 aspects of I).

However, if you make I structured, you
 can get a 2 fold increase in resolution

IF I is structured (not uniform) then
 ρ can be determined to a factor of 2
 better than normal diffraction limit

Analogy



see ripples

Get moiré fringes that are wider than
 diffraction limit

When you superimpose two ^{closely spaced} ~~grid~~ patterns on one another you get moiré fringes w/ a larger spacing

IF moiré fringes can be resolved by telescope (e.g. of spacing $> \sim 200 \text{ nm}$) knowledge of the fringes & knowledge of structured illumination allows determination of object to 2-fold better than diffraction

See picture

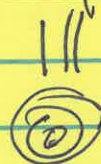
Collect more widely spaced moiré pattern

$$\begin{array}{ccc} \textcircled{F} & \propto & p \textcircled{I} \\ \downarrow \text{know} & & \downarrow \text{know} \end{array}$$

Pictures look bad but use computer to fix

Can VIDEO

two patterns



Do analogy to understand factor of 2.
Connects w/ MRI